

Synthetic mask generation using Stable Diffusion

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Abstract

During and after the COVID-19 pandemic, mask wearing became widespread, yet publicly available masked-face datasets remain limited and often lack diversity or proper annotation of correct versus incorrect usage. We propose a two-stage framework to synthesise realistic masked faces from clean portraits. Our approach first predicts the facial region corresponding to the mask and then leverages diffusion-based generation to produce realistic masked faces conditioned on this region. We evaluate two diffusion approaches: a denoising diffusion probabilistic model (DDPM) and a Stable Diffusion inpainting model fine-tuned with LoRA. Compared to a CycleGAN baseline, diffusion-based methods produce more realistic results and better preserve identity. In particular, LoRA-adapted Stable Diffusion achieves strong performance (FID 17.69, LPIPS 0.073, PSNR 23.56 dB), significantly outperforming both DDPM and GAN-based approaches. These results highlight the effectiveness of diffusion models for masked-face synthesis and suggest their potential for improving data augmentation in face-related computer vision tasks.

1. Introduction

Mask wearing became ubiquitous during the COVID-19 pandemic and remains common for medical, cultural or personal reasons. Although human vision easily recognises masked faces, computer-vision systems trained on unmasked faces often fail because the occlusion removes half of the facial features. Publicly available masked-face datasets are still scarce: researchers frequently create synthetic datasets by overlaying masks on existing facial images. Even then, existing masked datasets either lack diversity or do not distinguish between correctly and incorrectly worn masks. Consequently, there is a need for methods that can reliably synthesise realistic masked faces from clean portraits.

1.1. Motivation and problem definition

Real masked-face datasets are limited in size and variety, while collecting large facial datasets is costly and raises

privacy concerns. Synthetic generation can expand training data by increasing identity and mask-style diversity under controlled conditions. Our goal is to generate a realistic masked face from a clean portrait. This problem can be formulated as learning a conditional mapping G that, given an unmasked face $\mathbf{x}$ and a mask region m , synthesises a masked face $\mathbf{y} = G(\mathbf{x}, m)$ which preserves the subject’s identity and looks photo-realistic. Early works tackled related tasks using generative adversarial networks (GANs) [1]; however, GANs tend to overfit the training distribution and may degrade when test images differ in pose or alignment. We build upon diffusion models, which are likelihood-based generative models that produce high-quality images without adversarial training. We hypothesise that diffusion models can synthesise masks more reliably and generalise better to varied face poses.

1.2. Related work and Contributions

This paper makes the following contributions:

- We propose a two-stage synthesis pipeline for masked faces. First, a *U-Net mask predictor* learns to segment the region where a mask should be placed, given a clean face. Second, a *diffusion model* conditioned on the predicted mask generates the masked region while preserving identity. We investigate two variants: a denoising diffusion probabilistic model (DDPM) trained from scratch and a Stable Diffusion model fine-tuned with low-rank adapters (LoRA) [2]. By training only a small number of low-rank parameters, LoRA dramatically reduces the cost of adapting large pretrained diffusion model.
- We curate training data by combining 70k high-quality faces from FFHQ [3] with the MaskedFace-Net dataset [4], which provides 133k synthetic faces wearing masks correctly or incorrectly. The predicted masks guide the diffusion model to the desired region, allowing synthesis on out-of-distribution images.
- We benchmark our method against CycleGAN [5], the strongest GAN-based baseline for this task. Experiments show that our diffusion models achieve competitive FID scores and produce more identity-preserving results, while offering better flexibility at inference time.

The remainder of this report is organised as follows. Section 1.2 reviews related work on GAN-based mask gen-

eration and diffusion models. Section 2 describes the U-Net mask predictor and its training. Section 3 details our diffusion-based masked-face generators. Section 4 discusses fine-tuning Stable Diffusion with LoRA. Section 4.4 presents experimental results and analysis, and Section 5 concludes the paper.

2. U-Net Mask Predictor

In the first stage of our pipeline we learn to predict a binary mask that localises the region of the face where a protective covering should be placed. Accurate mask prediction is critical because it defines the editable region fed into the downstream diffusion model. We therefore train a dedicated U-Net [6] on pairs of clean faces and their corresponding ground-truth mask annotations. This section describes the dataset, architecture, training protocol, and evaluation metrics used for the mask predictor.

2.1. Dataset and preprocessing

We assemble a dataset of 2,000 training samples, 1,000 validation samples and 1,000 test samples as in the baseline setup. Each sample consists of a clean aligned face image and a binary mask that indicates the region where a medical mask should be generated. Images are converted to RGB and resized to 128×128 pixels using bilinear interpolation; corresponding mask images are resized using nearest-neighbour interpolation to preserve their binary nature. During training we apply random horizontal flips with probability 0.5 to both the clean face and mask to augment the data. We also create a binary mask from the ground-truth mask using binary threshold to use it during training.

2.2. Network architecture

We adopt a standard U-Net as our mask predictor. The network consists of an encoder that successively downsamples the input and a decoder that upsamples and merges features through skip connections. The network outputs a single-channel logits map of the same spatial dimensions as the input, which is later converted to a binary mask via a sigmoid and threshold operation.

2.3. Loss function and metrics

We train the U-Net using a composite loss that balances binary cross-entropy (BCE) with a soft-Dice loss. Concretely, for each mini-batch of logits $\ell \in \mathbb{R}^{B \times 1 \times 128 \times 128}$ and ground-truth masks $y \in \{0, 1\}^{B \times 1 \times 128 \times 128}$ we compute

$$\mathcal{L}_{\text{mask}} = 0.5 \mathcal{L}_{\text{BCE}}(\ell, y) + 0.5 \mathcal{L}_{\text{Dice}}(\ell, y), \quad (1)$$

Table 1. Hyperparameters and data statistics for U-Net mask predictor training.

Parameter	Value
Training samples	2,000
Validation samples	1,000
Image size	128×128
Batch size	8
Epochs	30
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Base channels	64
Dropout	0
Gradient clip	1.0
Random seed	7
Optimiser	AdamW
Loss	BCE + Dice

where

$$\mathcal{L}_{\text{BCE}}(\ell, y) = -\frac{1}{N} \sum_i [y_i \log \sigma(\ell_i) + (1 - y_i) \log(1 - \sigma(\ell_i))] \quad (2)$$

$$\mathcal{L}_{\text{Dice}}(\ell, y) = 1 - \frac{2 \sum_i \sigma(\ell_i) y_i + \varepsilon}{\sum_i \sigma(\ell_i) + \sum_i y_i + \varepsilon}, \quad (3)$$

and σ denotes the sigmoid function, $\varepsilon = 10^{-6}$ prevents division by zero, and the sums run over all spatial positions and mini-batch entries. The Dice term encourages overlap between predicted and ground-truth masks and alleviates class imbalance.

To evaluate mask quality we compute the intersection over union (IoU) and Dice score on the validation set. Given predicted binary mask $\hat{m}$ and ground-truth mask m , these metrics are defined as

$$\text{IoU}(\hat{m}, m) = \frac{|\hat{m} \cap m|}{|\hat{m} \cup m|}, \quad \text{Dice}(\hat{m}, m) = \frac{2|\hat{m} \cap m|}{|\hat{m}| + |m|}. \quad (4)$$

2.4. Training protocol

Training is carried out for 30 epochs with a batch size of 8. We use the AdamW optimiser with a learning rate of 10^{-4} and a weight decay of 10^{-4} to stabilise updates. Gradients are clipped to a maximum norm of 1.0 to prevent exploding gradients, and we employ PyTorch’s automatic mixed-precision to accelerate training on a single T4-GPU. A fixed random seed (7) ensures the reproducibility of results.

2.5. Qualitative results

Figure 1 illustrates the output of the mask predictor on several test images. For each example we display the input

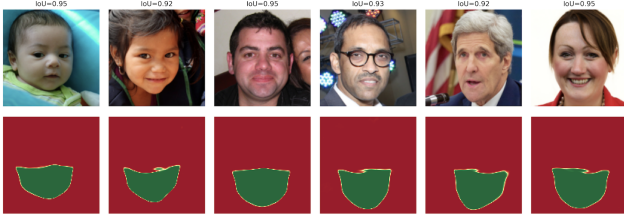


Figure 1. Qualitative examples of predicted masks. Each column shows (top) the input face along with the corresponding IoU score, and (bottom) the mask predicted by our U-Net. The model consistently identifies and localizes the lower facial region across diverse subjects, achieving high overlap scores (IoU $\approx$ 0.92–0.95), indicating accurate segmentation performance.

clean face and the corresponding prediction produced by our model. The predicted masks closely match the region of the mask and accurately delineate the region covering the nose and mouth, demonstrating that the U-Net has learned a precise localisation prior.

3. Diffusion Models for Mask Generation

Diffusion probabilistic models are a recent family of generative models that have achieved state-of-the-art sample quality by modelling the distribution of data as the reversal of a gradual noising process. In our second stage we employ a conditional denoising diffusion probabilistic model (DDPM) to synthesize a realistic masked face conditioned on the clean face and binary mask predicted in Section 2.

3.1. Overview of Denoising Diffusion Probabilistic Models

A diffusion model consists of two Markov chains: a fixed “forward” process that gradually adds Gaussian noise to data and a learned “reverse” process that removes the noise step by step. Let x_0 denote a clean data sample. The forward process defines a sequence $\{x_t\}_{t=1}^T$ through

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}), \quad (5)$$

where $(\beta_1, \dots, \beta_T)$ is a predefined variance schedule and T is the number of diffusion steps. This Markov chain converts the data into pure Gaussian noise as $t \rightarrow T$. Sampling from an arbitrary timestep can be performed analytically by accumulating the noise, and the probability of the entire forward chain factorizes as

$$q(x_{1:T} | x_0) = \prod_{t=1}^T \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}) \quad (6)$$

The reverse process is another Markov chain [7] with learned Gaussian transitions

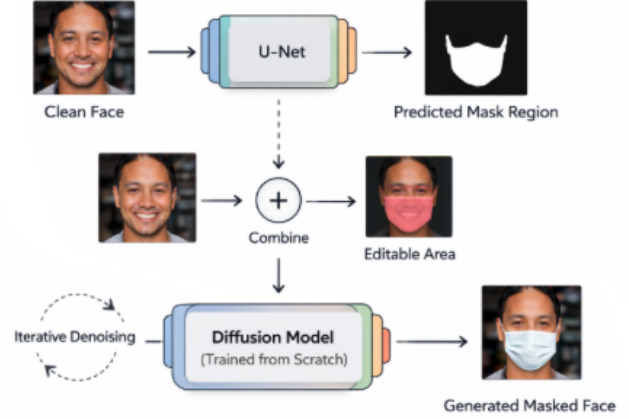


Figure 2. Pipeline of the proposed method. Given a clean face, a U-Net predicts the mask corresponding to the target region. The predicted mask is combined with the input image to define the editable area. A diffusion model then performs iterative denoising conditioned on this masked input to synthesize a realistic masked face.

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)), \quad (7)$$

where the means and variances are parameterised by a neural network.

3.2. Training a DDPM from scratch

We implement a conditional denoising diffusion model that predicts the masked face x_0 given a clean face image c and the corresponding binary mask m (both resized to 128×128). The network architecture is a U-Net with sinusoidal time embeddings and residual blocks, similar to the segmentation U-Net but augmented to accept conditioned inputs. In particular, the model concatenates the noisy image x_t with the condition tensor $u = [c \odot (1 - m), m]$ along the channel dimension. Our DDPM model is trained from scratch on the same dataset used for the mask predictor: 2,000 training images and 1,000 validation images. We set the batch size to 8 and train for 50 epochs. The optimiser is AdamW with learning rate 2×10^{-4} and weight decay 1×10^{-4} .

3.3. Noise schedule and parameter settings

The variance schedule β_t controls how quickly signal is destroyed in the forward process and therefore affects both training stability and sampling quality. We adopt the cosine schedule of [8], which defines the cumulative signal-to-noise ratio $\bar{\alpha}_t$ via a cosinesquared decay and then derives the per-step variances by normalisation:

$$\bar{\alpha}_t = \cos^2\left(\frac{(t/T)+s}{1+s} \frac{\pi}{2}\right), \quad \beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}, \quad (8)$$

where $T = 250$ is the total number of timesteps and $s = 0.008$ is a small offset to avoid numerical issues. In addition to the variance schedule, we follow the DDIM sampling strategy of [9] during inference. We generate 100 reverse diffusion steps instead of all T steps, which substantially reduces runtime while maintaining image quality.

3.4. Combining DDPM with the predicted mask

To generate a masked face from a clean face c we first predict the binary mask m using the U-Net mask predictor from Section 2. Let $c' = c \odot (1 - m)$ denote the clean image with the region under the mask removed (set to zero). We form a condition tensor $u = [c', m]$ by concatenating c' and m along the channel dimension. The diffusion model is trained to reconstruct the masked face within the mask region while preserving the clean pixels outside. After obtaining the final denoised output x_0 , we composite the generated mask region onto the original clean image to produce the final prediction. Figure 2 illustrates this pipeline and shows intermediate results during the denoising process.

In summary, the diffusion model produces realistic masked faces but, with limited training data, its FID remains higher than that of the CycleGAN baseline (See Table 3). Nevertheless, the ability to control the mask region via a binary mask offers greater flexibility and facilitates composition with the clean image.

4. Stable Diffusion Fine-Tuning

In addition to the DDPM trained from scratch, we study a second diffusion baseline based on a large pretrained latent diffusion model. We use the `runwayml/stable-diffusion-inpainting` checkpoint from Hugging Face Diffusers as the backbone and adapt it to the clean-to-masked face synthesis task with Low-Rank Adaptation (LoRA) [2]. This choice allows us to leverage a strong pretrained image prior while keeping the number of trainable parameters small.

Unlike the DDPM branch, which explicitly conditions on a predicted binary mask, the Stable Diffusion branch is formulated as full-image conditional diffusion. The conditioning input is the clean face image together with a text prompt describing the target domain, while the supervision target is the corresponding masked face image. In practice, the prompt used throughout training and inference is: “*a high quality portrait photo of a person wearing a protective face mask*”.

4.1. Frozen and trainable components

The tokenizer, CLIP text encoder, variational autoencoder (VAE), and the original pretrained U-Net weights are kept frozen. Adaptation is performed through broad LoRA modules inserted into the U-Net. We don’t do attention-only

LoRA, but we also apply LoRA to a broader subset of U-Net modules, including attention projections as well as additional projection and convolution layers. This gives the model more adaptation capacity while remaining significantly lighter than full U-Net fine-tuning.

4.2. Training setup

Training is performed on aligned FFHQ-CMFD image pairs at a spatial resolution of 256×256 . The model is optimized for 30 epochs with a per-device batch size of 2 and gradient accumulation of 4, giving an effective batch size of 8. We use AdamW with a constant learning rate of 10^{-4} .

The LoRA configuration uses rank 16 and scaling factor $\alpha = 16$. During training, the target masked image is encoded into the latent space of the pretrained VAE, Gaussian noise is added according to the diffusion schedule, and the LoRA-adapted U-Net is trained to predict the added noise conditioned on the clean face latent and the prompt embedding.

4.3. Checkpoint selection and inference

All reported Stable Diffusion results use the final checkpoint obtained after 30 epochs of training. At inference time, we use 30 denoising steps and a classifier-free guidance scale of 7.5.

The pretrained backbone used in all experiments is therefore: `runwayml/stable-diffusion-inpainting`, and the final task-specific adaptation is provided by the learned LoRA weights.

4.4. Experimental evaluation

We evaluate the Stable Diffusion branch using both distribution-level and paired-image metrics. FID measures the discrepancy between generated and real masked-face distributions, while LPIPS, PSNR, and MAE-based metrics measure similarity with respect to the paired ground-truth masked faces.

Table 2. Evaluation results for the Stable Diffusion branch. Lower is better for FID, LPIPS, MAE-all, MAE-mask, and MAE-background, while higher is better for PSNR.

Metric	Value
FID ↓	17.69
LPIPS ↓	0.073
PSNR ↑	23.56
MAE-all ↓	0.0362
MAE-mask ↓	0.0802
MAE-background ↓	0.0268

Table 3 reports the FID values obtained by the three gen-

erative approaches considered in this project. These values provide a useful empirical comparison, but they are not strictly directly comparable because the methods were not evaluated on exactly the same held-out samples.

Table 3. FID comparison across generative methods. Lower is better. (StableDiffusion Train FID isn’t shown because of its different training set size)

Method	Epochs	Train FID ↓	Test FID ↓
CycleGAN	30	28.43	28.48
DDPM	50	47.09	48.93
Stable Diffusion	30	–	17.69

The Stable Diffusion results show strong perceptual quality, with low LPIPS and MAE values together with a relatively high PSNR. In addition, the FID obtained by the Stable Diffusion branch is lower than the values observed for the DDPM and CycleGAN baselines in our experiments. Nevertheless, since the evaluation subsets are not exactly the same across methods, this comparison should be interpreted as indicative rather than as a perfectly controlled benchmark.



Figure 3. Comparison of masked-face synthesis results across different methods. The Stable Diffusion approach produces more realistic masks and better preserves facial identity.

5. Conclusion

This work investigated diffusion-based approaches for synthetic face-mask generation from clean portraits. We explored two complementary directions: a two-stage DDPM pipeline based on mask prediction followed by conditional diffusion, and a Stable Diffusion baseline adapted through broad LoRA fine-tuning. Together, these experiments show that diffusion models provide a flexible and effective framework for realistic masked-face synthesis.

The Stable Diffusion model achieved an FID of 17.7, showing that parameter-efficient adaptation of a pretrained diffusion model can produce realistic masked-face synthesis at relatively moderate training cost. More broadly, the

project confirms that diffusion-based methods are a promising direction for masked-face generation, both when training a model from scratch and when adapting a large pre-trained generator.

A main limitation observed in the Stable Diffusion results is that some generated faces remain visually distorted even when the global metrics are favorable. This suggests that FID, PSNR, LPIPS, and MAE are useful but not sufficient on their own to fully capture facial realism and identity preservation. In this context, a safer direction is to use an inpainting pipeline conditioned on an explicit mask, since constraining the generation to the mask region is likely to be more robust and to reduce the risk of altering the rest of the face. Future work should therefore combine stronger mask-conditioned generation with evaluation protocols that better reflect identity preservation and local facial quality.

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